Operating Systems Notes - CPU Scheduling

Introduction

CPU Scheduling is the process used by the Operating System to decide which process gets access to the CPU at a particular time. Since many processes may be waiting for execution, the scheduler selects one process from the ready queue.

Objectives of CPU Scheduling

• Maximize CPU utilization
• Increase throughput
• Reduce waiting time
• Reduce turnaround time
• Reduce response time
• Ensure fairness among processes

Basic Terms

Arrival Time (AT): Time at which a process enters the ready queue.
Burst Time (BT): Time required by a process for CPU execution.
Completion Time (CT): Time at which process finishes execution.
Turnaround Time (TAT): CT - AT
Waiting Time (WT): TAT - BT
Response Time (RT): Time from arrival until first response.

Types of Scheduling

Preemptive Scheduling: CPU can be taken away from a running process.
Example: Round Robin, SRTF.

Non-Preemptive Scheduling: Once CPU is assigned, process keeps CPU until completion.
Example: FCFS, SJF.

First Come First Serve (FCFS)

• Processes are executed in the order of arrival.
• Simplest scheduling algorithm.
• Non-preemptive.
• Can cause Convoy Effect where short processes wait for long ones.

Shortest Job First (SJF)

• Process with shortest burst time executes first.
• Minimizes average waiting time.
• Can be preemptive or non-preemptive.
• Difficult to predict exact burst time.

Shortest Remaining Time First (SRTF)

• Preemptive version of SJF.
• Process with smallest remaining burst time gets CPU.
• Better response time than SJF.
• Higher overhead due to frequent context switching.

Priority Scheduling

• CPU allocated based on process priority.
• Higher priority process executes first.
• Can be preemptive or non-preemptive.
• May cause starvation of low priority processes.
• Aging technique helps avoid starvation.

Round Robin (RR)

• Each process gets fixed time quantum.
• After quantum expires, process moves to back of queue.
• Widely used in time-sharing systems.
• Very fair scheduling algorithm.

Multilevel Queue Scheduling

• Ready queue divided into multiple queues.
• Each queue may have different scheduling algorithm.
• Processes permanently assigned to queues.

Multilevel Feedback Queue

• Processes can move between queues.
• Provides flexibility and better performance.
• Prevents starvation using dynamic priorities.

Context Switching

Context switching is the process of saving the state of one process and loading the state of another process. Excessive context switching reduces CPU efficiency.

Dispatcher

Dispatcher is the module that gives control of CPU to the selected process. Dispatcher latency is the time taken to stop one process and start another.

Comparison of Scheduling Algorithms

Algorithm	Type	Advantages	Disadvantages
FCFS	Non-Preemptive	Simple and easy	High waiting time
SJF	Both	Minimum waiting time	Burst prediction difficult
SRTF	Preemptive	Good response time	More context switches
Priority	Both	Important tasks execute first	Starvation possible
Round Robin	Preemptive	Fair and responsive	Context switch overhead

Conclusion
CPU Scheduling is one of the most important responsibilities of an Operating System. Different scheduling algorithms are used depending on system requirements such as fairness, responsiveness, and throughput.